

DC

Ex 1.

$$m\ddot{d} = f - c\dot{d} - kd$$
$$\Rightarrow \ddot{d} = -\frac{c}{m}\dot{d} - \frac{k}{m}d + \frac{f}{m}$$

$$\begin{cases} x_1 = d \\ x_2 = \dot{d} \end{cases} \quad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\Rightarrow \ddot{x}_2 = -\frac{c}{m}x_2 - \frac{k}{m}x_1 + \frac{f}{m}$$

$$\Rightarrow \dot{\underline{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{c}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/m \end{bmatrix} f$$

$$\underline{y} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix} \underline{x}$$

Ex 2. $G(s) = \frac{Y(s)}{U(s)} = \frac{1}{s^2}$

$$\Rightarrow s^2 Y(s) = U(s)$$

$$\Rightarrow \ddot{y}(t) = u(t)$$

Let $x_1 = y$ $x_2 = \dot{y}$, then $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$

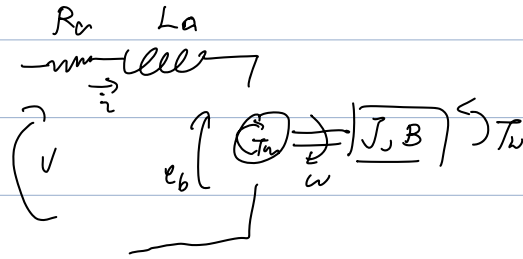
$$\Rightarrow \dot{x}_2 = u(t)$$

$$\text{then } \dot{\underline{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\underline{F}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{\underline{G}} u(t)$$

$$\underline{y} = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\underline{H}} \underline{x}$$

Ex3.

$$\left\{ \begin{array}{l} T_m = K_T i \\ e_b = K_b \omega \\ V = i R_a + L_a \dot{i} - e_b \\ J \dot{\omega} = T_m - B \omega \end{array} \right.$$



$$\Rightarrow \left\{ \begin{array}{l} V = i R_a + L_a \dot{i} - K_b \omega \\ J \dot{\omega} = K_T i - B \omega \end{array} \right.$$

LAPLACE

$$\Rightarrow \left\{ \begin{array}{l} (L_a s + R_a) I(s) = V(s) - K_b \Omega(s) \\ (J s + B) \Omega(s) = K_T I(s) \end{array} \right.$$

$$\Rightarrow (L_a s + R_a) \cdot \frac{1}{K_T} (J s + B) \Omega(s) = V(s) - K_b \Omega(s)$$

$$\Rightarrow \left[\frac{1}{K_T} (L_a J s^2 + B L_a s + R_a J s + R_a B) + K_b \right] \Omega(s) = V(s)$$

$$\Rightarrow \left(\frac{L_a J}{K_T} s^2 + \frac{B L_a + R_a J}{K_T} s + \frac{R_a B}{K_T} + K_b \right) \Omega(s) = V(s)$$

$$\Rightarrow \frac{\Omega(s)}{V(s)} = \frac{K_T}{L_a J s^2 + (B L_a + R_a J) s + R_a B + K_T K_b}$$

When $L_a \rightarrow 0$, then $\frac{\Omega(s)}{V(s)} = \frac{K_T}{R_a J s + R_a B + K_T K_b}$

EX4,

$$D(s) = \frac{U(s)}{E(s)} = K \frac{s+a}{s+b}$$

$$\Rightarrow U(s) \cdot (s+b) = K (s+a) \cdot E(s)$$

$$\Rightarrow u(t) + b u(t) = K \dot{e}(t) + K a e(t)$$

$$\Rightarrow \frac{u(k+1) - u(k)}{T} + b u(k) = K \left(\frac{e(k+1) - e(k)}{T} + a e(k) \right)$$

$$\Rightarrow u(k+1) = u(k) + T(\dots)$$

EX5. $u(t) = K e(t) + \frac{K}{T_I} \int_0^t e(\tau) d\tau + K T_D \dot{e}(t)$

$$\Rightarrow \dot{u}(t) = K \dot{e}(t) + \frac{K}{T_I} e(t) + K T_D \ddot{e}(t)$$

$$\Rightarrow \frac{u(k+1) - u(k)}{T} = K \frac{e(k+1) - e(k)}{T} + \frac{K}{T_I} e(k) + K T_D \frac{\dot{e}(k+1) - \dot{e}(k)}{T}$$

$$\frac{e(k+1) - e(k)}{T} - \frac{e(k+1) - e(k)}{T}$$

$$\frac{e(k+2) - 2e(k+1) + e(k)}{T^2}$$

EX6/7 $u_k = u_{k-1} + u_{k-2} \quad \begin{cases} u_0 = 1 \\ u_1 = 1 \end{cases}$

Let $u_k = A z^k$

then, $A z^k = A z^{k-1} + A z^{k-2} \quad (A \neq 0, z \neq 0)$

$$\Rightarrow z^2 = z + 1$$

$$z_{1,2} = \frac{1}{2} \pm \frac{\sqrt{5}}{2}$$

SUBBING:

$$u_k = A_1 z_1^k + A_2 z_2^k$$

$$k=0; \quad u_0 = A_1 + A_2 = 1$$

$$k=1: \quad u_1 = A_1 z_1 + A_2 z_2 = 1$$

$$\Rightarrow A_1 = \frac{\sqrt{5}+1}{2\sqrt{5}} \quad A_2 = \frac{\sqrt{5}-1}{2\sqrt{5}}$$

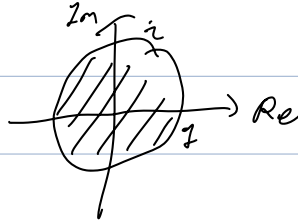
$$A_1 z_1 + z_2 - A_1 z_1 = 1$$

$$\Rightarrow A_1 = \frac{1-z_2}{z_1-z_2} = \frac{\frac{1}{2} + \frac{\sqrt{5}}{2}}{\frac{\sqrt{5}}{2}} = \frac{\sqrt{5}+1}{2\sqrt{5}}$$

Closed form solution:

$$u_k = \frac{\sqrt{5}+1}{2\sqrt{5}} \left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)^k + \frac{\sqrt{5}-1}{2\sqrt{5}} \left(\frac{1}{2} - \frac{\sqrt{5}}{2}\right)^k$$

Stable if: $|z_1| < 1$ And $|z_2| < 1$



Ex 8: $u_k = 0.9u_{k-1} - 0.2u_{k-2}$

$$\Rightarrow z^2 = 0.9z - 0.2$$

$$\Rightarrow z^2 - 0.9z + 0.2 = 0$$

$$\Rightarrow z_{1,2} = 0.45 \pm \sqrt{0.45^2 - 0.2}$$

$$|z_{1,2}| < 1 \quad \text{stable.}$$

Ex 14. $F(z) = \frac{2z^2 + 5z}{6z^2 + 7z - 20}$

$$\begin{array}{r} \frac{1}{3} + \frac{4}{3}z + \dots \\ 6z^2 + 7z - 20 \overline{) 2z^2 + 5z} \\ \underline{2z^2 + \frac{7}{3}z - \frac{20}{3}} \\ \frac{8}{3}z + \frac{20}{3} \end{array}$$

Ex 15. $F(z) = \frac{z(7z-23)}{(z-2)(z-5)} = \frac{A}{z-2} + \frac{B}{z-5} = \frac{(A+B)z - 5A - 2B}{(z-2)(z-5)}$

$$\Rightarrow \begin{cases} A+B=7 \\ 5A+2B=23 \end{cases} \Rightarrow \begin{cases} A=3 \\ B=4 \end{cases}$$

$$\Rightarrow \frac{F(z)}{z} = \frac{3}{z-2} + \frac{4}{z-5}$$

$$\Rightarrow F(z) = 3 \frac{\frac{z}{2}}{\frac{z}{2} - 1} + 4 \frac{\frac{z}{5}}{\frac{z}{5} - 1}$$

$$\Rightarrow f_k = 3 \cdot 2^k \mathbb{1}(kT) + 4 \cdot 5^k \mathbb{1}(kT), \quad k \geq 0$$

$$= 3 \cdot 2^k + 4 \cdot 5^k, \quad k \geq 0,$$

Ex 16. $F(z) = \frac{-z^2 + 2z + 18}{(z-3)^2}$

$$\Rightarrow F(z)/z = \frac{-z^2 + 2z + 18}{z(z-3)^2} = \frac{A}{z} + \frac{Bz+C}{(z-3)^2} + \frac{D}{z-3}$$

$$= \frac{A(z-3)^2 + z(Bz+C) + Dz(z-3)}{z(z-3)^2}$$

$$= \frac{(A+B+D)z^2 + (-6A+C-3D)z + 9A}{z(z-3)^2}$$

$$\Rightarrow \begin{cases} A+B+D = -1 \\ -6A+C-3D = 2 \\ 9A = 18 \end{cases} \Rightarrow \begin{cases} A = 2 \\ B = 0 \\ C = 5 \\ D = -3 \end{cases}$$

$$\Rightarrow F(z)/z = \frac{2}{z} + \frac{5}{(z-3)^2} + \frac{-3}{z-3}$$

$$\Rightarrow F(z) = 2 + 5 \frac{z}{(z-3)^2} - 3 \frac{z}{z-3}$$

Ex 17. $y_{k+2} = y_{k+1} + y_k$

$$\Rightarrow z^2 Y(z) - z^2 y_0 - z y_1 = z Y(z) - z y_0 + Y(z)$$

$$\Rightarrow z^2 Y(z) - z^2 - z = z Y(z) - z + Y(z)$$

$$\Rightarrow Y(z) = \frac{z^2}{z^2 - z - 1}$$

$$Y(z)/z = \frac{z}{z^2 - z - 1} = \frac{A}{z-a} + \frac{B}{z-b}$$

$$a = \frac{1}{2} + \sqrt{\frac{1}{4} + 1} = \frac{1}{2} + \frac{\sqrt{5}}{2}$$

$$b = \frac{1}{2} - \sqrt{\frac{1}{4} + 1} = \frac{1}{2} - \frac{\sqrt{5}}{2}$$

$$\begin{cases} A+B=1 \\ -bA-aB=0 \end{cases}$$

$$\Rightarrow \begin{cases} A = \dots \\ B = \dots \end{cases}$$

$$\Rightarrow Y(z) = \dots$$

$$\Rightarrow y_k = \dots$$

Ex 16

$$Y(z) = 6[z^{-1}Y(z) + y_{-1}] + 4[z^{-2}Y(z) + z^{-1}y_1 + y_2]_{z=0}$$

Ex 19. $e_k = kT 1(k)$

$$E(z) = \sum_{k=-\infty}^{\infty} kT 1(k) \cdot z^k$$

$$= Tz^{-1} + 2Tz^{-2} + 3Tz^{-3} + \dots$$

$$= Tz^{-1} (1 + 2z^{-1} + 3z^{-2} + \dots)$$

$$= Tz^{-1} (1 - z^{-1})^{-2}$$

$$= T \frac{z^{-1}}{(1 - z^{-1})^2} = \frac{zT}{(z-1)^2}$$

$$f = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3} + \dots$$

$$z^{-1}f = z^{-1} + 2z^{-2} + 3z^{-3} + \dots$$

$$f - z^{-1}f = 1 + z^{-1} + z^{-2} + \dots = \frac{1}{1 - z^{-1}}$$

$$\Rightarrow f = \frac{1}{(1 - z^{-1})^2}$$

Ex 20.

$$y(k+1) - y(k) = u(k+1) + u(k)$$

$$\Rightarrow zY(z) - y_0 - Y(z) = zU(z) - y_0 + U(z)$$

$$\Rightarrow \frac{Y(z)}{U(z)} = \frac{z+1}{z-1}$$

$$\text{Let } U(z) = \frac{zT}{(z-1)^2}$$

$$\text{then, } Y(z) = \frac{z+1}{z-1} \cdot \frac{zT}{(z-1)^2} = \frac{(z+1)zT}{(z-1)^3}$$

$$y(k) = \frac{1}{T} \cdot (kT)^2 = k^2 T \quad (k \geq 0)$$

proof: $x(k) = (kT)^2$

$$X(z) = \sum_{k=-\infty}^{\infty} x(k) \cdot z^{-k}$$

$$a_k = k^2 e^{-k}$$

$$= T^2 \sum k^2 z^{-k}$$

$$S_k - S_{k-1} = k^2 e^{-k}$$

$$= T^2 \cdot \frac{z(z+1)}{(z-1)^3}$$

$(e \rightarrow z)$ $\dot{\hat{z}} \hat{z} k!$

$$S_k = e^{-1} + 2^2 \cdot e^{-2} + 3^2 \cdot e^{-3} + \dots + k^2 \cdot e^{-k}$$

z域微分

$$e S_k = 1 + 2^2 \cdot e^{-1} + 3^2 \cdot e^{-2} + 4^2 \cdot e^{-3} + \dots + k^2 \cdot e^{-k+1}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$(e-1) S_k = 1 + (2^2-1)e^{-1} + (3^2-2^2)e^{-2} + (4^2-3^2)e^{-3} + \dots$$

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} (-n) x(n) \cdot z^{-n-1}$$

$$+ (k^2 - (k-1)^2) e^{-k+1}$$

$$= k^2 e^{-k}$$

$$\Rightarrow -z \cdot \frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} n x(n) \cdot z^{-n}$$

$$= 1 + 3e^{-1} + 5e^{-2} + 7e^{-3} + \dots + (3-2k)e^{-k+2}$$

$$= k^2 e^{-k}$$

$$= Z[nx(n)]$$

$$e(e-1) S_k = e + 3 + 5e^{-1} + 7e^{-2} + 9e^{-3} + \dots + (3-2k) \cdot e^{-k+2}$$

$$= k^2 e^{-k+1}$$

$$(e-1)^2 S_k = e+2 + 2(e^{-1} + e^{-2} + e^{-3} + \dots + e^{-k+2}) - k^2 \cdot e^{-k+1} + k^2 \cdot e^{-k}$$

$$= e+2 + 2\left(\frac{1-\frac{1}{e^k}}{1-\frac{1}{e}}\right) \cdot 1 + k^2 e^{-k}(1-e)$$

$$= e + \frac{2e}{e-1} + k^2 e^{-k}(1-e)$$

$$= \frac{e(e+1) - k^2 e^{-k}(e-1)^2}{e-1}$$

$$\Rightarrow S_{\infty} = \frac{e(e+1)}{(e-1)^3}$$

Ex 21. $H = \frac{1}{(z-0.2)(z-0.4)}$

$$u_R = 5 \cos(0.3k + 0.1)$$